The Exact Finite-n σ-MGF for A_n

Proof, orbit splitting, and matrix verification

Computational proof companion

17 July 2026

Abstract

For $n \geq 2$, this note proves the exact cycle-type formula for the permutation representation of A_n, derives the correction to the S_n vector-partition count from orbit splitting, and proves equality with S_n in degree at most $n-2$. Explicit even permutation matrices and Reynolds projectors verify the formula, including a boundary case with a nonzero splitting correction.

1 From S_n to A_n

For any class function F on S_n, the indicator of an even permutation gives

$$\mathbb{E}_{A_n}[F] = \frac{1}{n!} \sum_{g \in S_n} (1 + \operatorname{sgn}(g)) F(g).$$

As in the symmetric-group calculation, the coefficient of m_τ is the trace, hence the rank, of the Reynolds projector on $W_\tau = \bigotimes_b \operatorname{Sym}^{\tau_b}(\mathbb{C}^n)$.

If $\lambda \vdash n$, then

$$\operatorname{sgn}(\lambda) = (-1)^{n-\ell(\lambda)}, \qquad z_\lambda = \prod_{d \geq 1} d^{m_d(\lambda)} m_d(\lambda)!.$$

Combining the sign indicator with the determinant of each permutation cycle proves the exact expression

$$\boxed{\mathcal{M}_{A_n} = \sum_{\lambda \vdash n} \frac{1 + (-1)^{n-\ell(\lambda)}}{z_\lambda} \prod_i \prod_{d \geq 1} (1 - t_i^d)^{-m_d(\lambda)}.}$$

Only even cycle types remain, each with coefficient $2/z_\lambda$.

2 Generating function

Set

$$G_+(u) = \prod_{\mathbf{v} \geq 0} (1 - ut^{\mathbf{v}})^{-1}, \qquad G_-(u) = \prod_{\mathbf{v} \geq 0} (1 + ut^{\mathbf{v}}).$$

The first product records multisets of labels. The second records sets of distinct labels and is the sign-weighted cycle-index contribution. Hence, for $n \geq 2$,

$$\boxed{\mathcal{M}_{A_n} = [u^n]\big(G_+(u) + G_-(u)\big).}$$

Equivalently, applying the exponential formula to both factors gives

$$\mathcal{M}_{A_n} = [u^n]\Bigg[\exp\left(\sum_{d \geq 1} \frac{u^d}{d} \prod_i \frac{1}{1 - t_i^d}\right) + \exp\left(\sum_{d \geq 1} \frac{(-1)^{d-1} u^d}{d} \prod_i \frac{1}{1 - t_i^d}\right)\Bigg].$$

3 Exact orbit interpretation

An S_n-orbit of monomial-basis labelings is determined by multiplicities $(a_{\mathbf{v}})_{\mathbf{v}\geq 0}$. Its stabilizer is $\prod_{\mathbf{v}} S_{a_{\mathbf{v}}}$. If some $a_{\mathbf{v}} \geq 2$, the stabilizer contains the odd transposition of two equally labeled points, so the S_n-orbit remains a single A_n-orbit. If every $a_{\mathbf{v}}$ is zero or one, the stabilizer lies in A_n, and the orbit splits into two A_n-orbits. Thus

$$[m_\tau]\mathcal{M}_{A_n} = \#\left\{(a_{\mathbf{v}}) \geq 0 : \sum a_{\mathbf{v}} = n,\ \sum a_{\mathbf{v}}\mathbf{v} = \tau\right\} + \#\left\{(a_{\mathbf{v}}) \in \{0,1\} : \sum a_{\mathbf{v}} = n,\ \sum a_{\mathbf{v}}\mathbf{v} = \tau\right\}.$$

The first term is the S_n coefficient. The correction counts sets of distinct nonzero vectors summing to τ, with either n such vectors or $n-1$ such vectors and one zero vector.

4 Stable agreement with S_n

Any labeling of total degree $|\tau|$ has at most $|\tau|$ nonzero labels. If $|\tau| \leq n-2$, at least two labels are zero. Their transposition is odd and lies in the stabilizer, so no S_n-orbit can split. Therefore

$$\boxed{[m_\tau]\mathcal{M}_{A_n} = [m_\tau]\mathcal{M}_{S_n} \quad \text{whenever } |\tau| \leq n-2.}$$

Outside that range the distinct-label term is the exact correction, not merely an error estimate.

5 Verification method and evidence

The notebook enumerates even permutations, constructs their permutation matrices, builds the full matrices on each symmetric tensor block, and averages them to obtain the A_n Reynolds projector. Its rank is compared independently with the character average, the even-cycle-type formula, and the S_n orbit count plus a distinct-label dynamic-programming correction.

Case	$\dim W_\tau$	rank	cycle	S_n	corr.	total
$A_3, \tau = (2)$	6	2	2	2	0	2
$A_3, \tau = (3)$	10	4	4	3	1	4
$A_4, \tau = (2,1)$	40	4	4	4	0	4
$A_5, \tau = (2,2)$	225	9	9	9	0	9
$A_5, \tau = (3,1)$	175	7	7	7	0	7

The $A_3, \tau = (3)$ row deliberately lies outside the stable range and confirms that one S_3 orbit splits, increasing the invariant dimension from three to four. The remaining rows test multiblock targets and larger projectors. Idempotence residuals are at most 1.1×10^{-15}.

Conclusion. The sign-weighted cycle formula and orbit-splitting formula agree exactly, and both match direct matrix invariant dimensions in all representative cases.